

The Geometric Part of Decoherence: Quasi-Orthogonality in High-Dimensional Hilbert Spaces

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(Dated: May 26, 2026)

We isolate a geometric mechanism that complements the dynamical suppression of macroscopic interference: In a high-dimensional Hilbert space, almost all state vectors are nearly orthogonal, accommodating an exponentially large reservoir of mutually quasi-orthogonal environmental records. This geometry explains why macroscopic alternatives fail to exhibit visible interference once such records are populated. The argument is conditional and finite-dimensional, and it leaves the interpretive core of quantum mechanics untouched: geometry alone does not select a pointer basis, does not guarantee that a given Hamiltonian drives the system into typical regions of the accessible subspace, and does not turn an improper mixture into a proper one. It merely supplies the vast Hilbert-space capacity that makes decoherence so overwhelmingly effective for all practical purposes.

I. INTRODUCTION

The quantum measurement problem asks how the continuous and reversible Schrödinger evolution gives rise to the discrete, irreversible classical world we observe [1, 2]. Operationally, the question reduces to why the off-diagonal interference terms of a system’s reduced density matrix are unobservable on macroscopic scales.

The urgency of this question was forcefully articulated by Schrödinger in a 1952 colloquium [3]. He noted that if the wave equation is universally valid, then alternative macroscopic outcomes must all “really happen simultaneously.” Schrödinger argued that if nature actually took this form, we should find our surroundings rapidly blurring together into a “quagmire, or sort of a featureless jelly”—a variant of his cat metaphor. For example, a watch left unobserved in a drawer overnight should dissolve into a smeared probability distribution. This “jellification” argument poses a sharp, physical question: if unitary quantum mechanics holds at all scales, what fundamental mechanism prevents the simultaneous alternatives of the macroscopic world from blurring into an unobservable, featureless smear?

Bell coined “FAPP” (*for all practical purposes*) to express skepticism toward explanations that rest on the practical impossibility of observing macroscopic interference rather than on a fundamental physical mechanism [4]. Modern decoherence theory [1, 5–7] answers this by showing that interaction with a macroscopic environment dynamically drives the off-diagonal terms toward zero in a preferred (pointer) basis selected by the system–environment coupling.

We shall examine a complementary, purely *geometric* contribution to the same suppression. The relevant tools are the exponential packing of approximately orthogonal unit vectors and the concentration of measure on high-dimensional spheres [8–12]. These results are standard in

high-dimensional geometry, but their direct application to the environmental records appearing in decoherence is, to our knowledge, underemphasized.

Our central claim is modest. High-dimensional Hilbert-space geometry guarantees an enormous reservoir of nearly non-interfering pure states. This is a kinematic fact independent of any Hamiltonian. Whether actual environmental branches occupy this reservoir is a separate dynamical question. In ordinary decoherence theory, the system–environment interaction selects a pointer basis and correlates different pointer values with different environmental records. The geometric estimates below apply once those records are sufficiently typical in the accessible environmental subspace.

We therefore separate two ingredients that are often bundled together. The dynamical ingredient is the production of environmental records by the Hamiltonian. The kinematic ingredient is the concentration of measure on high-dimensional spheres, which makes generic records almost mutually orthogonal. The latter sharply quantifies the FAPP suppression of macroscopic interference, while leaving untouched the deeper interpretive question of whether an improper mixture has become a proper one.

II. RANDOM QUANTUM STATES, TYPICAL OVERLAPS, AND EXPONENTIAL PACKING

Heuristic argument

The main geometric point can be understood before any probability formalism is introduced. A normalized quantum state

$$|\psi\rangle = (\psi_1, \dots, \psi_d) \in \mathbb{C}^d \quad (1)$$

satisfies

$$|\psi_1|^2 + \dots + |\psi_d|^2 = 1. \quad (2)$$

If the state is chosen without any preferred direction, this unit probability weight is spread over d complex co-

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ordinates. Thus a typical coordinate weight should be of order $1/d$. Now fix a pure state $|\phi\rangle$. By unitary invariance one may choose coordinates so that $|\phi\rangle = |e_1\rangle$. The Born probability for a system prepared in $|\psi\rangle$ to pass the yes-no test associated with $|\phi\rangle$ is then

$$|\langle\phi|\psi\rangle|^2 = |\psi_1|^2. \quad (3)$$

Hence the transition probability from a random pure state to a fixed pure state is expected to be of order $1/d$. A transition probability larger than some fixed threshold $\varepsilon > 0$ would require one coordinate to carry an anomalously large fraction of the total norm. In high dimension this is exponentially unlikely. The exact calculation below makes this heuristic precise:

$$\mathbb{P}\{|\langle\phi|\psi\rangle|^2 \geq \varepsilon\} = (1 - \varepsilon)^{d-1} \leq \exp[-(d-1)\varepsilon]. \quad (4)$$

This exact overlap tail is the source of both typical quasi-orthogonality and the exponential packing estimate used below.

Notation and elementary distributions

The symbol $\mathbb{P}$ denotes the probability of an event, and $\mathbb{E}$ denotes expectation with respect to the random choice of state vectors. A Haar-random unit vector in $\mathbb{C}^d$ means a unit vector distributed according to the unique unitarily invariant probability measure on the unit sphere: applying any unitary $U \in U(d)$ leaves its distribution unchanged. In the real comparison below, Haar-random means uniformly distributed with respect to the orthogonally invariant probability measure on the real sphere $S^{d-1} \subset \mathbb{R}^d$.

The notation $N(0, 1)$ denotes the real normal distribution with mean zero and variance one, with density

$$(2\pi)^{-1/2} \exp(-x^2/2), \quad x \in \mathbb{R}. \quad (5)$$

Thus $X, Y \sim N(0, 1)$ means that X and Y are independent standard real Gaussian random variables. A standard complex Gaussian variable is

$$Z = \frac{X + iY}{\sqrt{2}}. \quad (6)$$

The factor $1/\sqrt{2}$ normalizes the second moment:

$$\mathbb{E}|Z|^2 = \frac{\mathbb{E}X^2 + \mathbb{E}Y^2}{2} = 1. \quad (7)$$

Equivalently, Z has density $\pi^{-1}e^{-|z|^2}$ on the complex plane. A standard complex Gaussian vector $Z = (Z_1, \dots, Z_d) \in \mathbb{C}^d$ has independent standard complex Gaussian components.

We shall use a few elementary distributional facts. A chi-square random variable with k degrees of freedom, denoted χ_k^2 , is the sum of the squares of k independent

$N(0, 1)$ variables. Hence, if $Z = (X + iY)/\sqrt{2}$ is a standard complex Gaussian, then

$$|Z|^2 = \frac{X^2 + Y^2}{2}, \quad X^2 + Y^2 \sim \chi_2^2. \quad (8)$$

Geometrically, $X^2 + Y^2$ is the squared radius of a two-dimensional isotropic Gaussian point. Its density is

$$\frac{1}{2}e^{-s/2}, \quad s \geq 0, \quad (9)$$

so $(X^2 + Y^2)/2$ has density e^{-t} on $t \geq 0$, i.e. the exponential distribution of mean one. We write this as

$$|Z|^2 \sim \text{Exp}(1) = \text{Gamma}(1, 1). \quad (10)$$

For different coordinates j , the variables $|Z_j|^2$ are independent because the Gaussian components Z_j are independent and $|Z_j|^2$ depends only on the j th component.

We use the shape-scale convention for gamma distributions: $\text{Gamma}(\alpha, \theta)$ has density proportional to $x^{\alpha-1}e^{-x/\theta}$ on $x > 0$, with shape α and scale θ . Thus $\text{Gamma}(\alpha, 1)$ has unit scale. The beta distribution $\text{Beta}(a, b)$ has density

$$\frac{1}{B(a, b)} t^{a-1} (1-t)^{b-1}, \quad 0 < t < 1, \quad (11)$$

where $B(a, b)$ is the beta function, i.e. the normalizing integral. We use the standard beta-ratio fact: if $U \sim \text{Gamma}(a, \theta)$ and $V \sim \text{Gamma}(b, \theta)$ are independent with the same scale θ , then

$$\frac{U}{U+V} \sim \text{Beta}(a, b). \quad (12)$$

The scale parameter cancels in the ratio.

Exact complex overlap distribution

The point is directly quantum-mechanical. If a system is prepared in $|\psi\rangle$ and tested against the pure state $|\phi\rangle$, the Born probability of the corresponding yes-outcome is

$$\langle\psi|(|\phi\rangle\langle\phi|)|\psi\rangle = |\langle\phi|\psi\rangle|^2. \quad (13)$$

Thus quasi-orthogonality of random pure states is equivalently a statement about the typical smallness of transition probabilities.

Lemma 1 (Complex random transition probabilities). *Let $|\psi\rangle, |\phi\rangle \in \mathbb{C}^d$ be independently chosen Haar-random unit vectors, with $d \geq 2$. Then, for $0 \leq \varepsilon \leq 1$,*

$$\mathbb{P}\{|\langle\phi|\psi\rangle|^2 \geq \varepsilon\} = (1 - \varepsilon)^{d-1}. \quad (14)$$

Equivalently, for $0 \leq \delta \leq 1$,

$$\mathbb{P}\{|\langle\phi|\psi\rangle| \geq \delta\} = (1 - \delta^2)^{d-1} \leq \exp[-(d-1)\delta^2]. \quad (15)$$

Proof. By unitary invariance of Haar measure, we may fix $|\phi\rangle = |e_1\rangle$, the first vector of an orthonormal basis. Indeed, conditioning on any value of $|\phi\rangle$, a unitary sending that vector to $|e_1\rangle$ leaves the distribution of the independently chosen $|\psi\rangle$ Haar-random. Thus averaging over two independent random vectors has the same overlap distribution as fixing one vector and averaging over the other. It remains to determine the distribution of $|\psi_1|^2$, where $|\psi\rangle = (\psi_1, \dots, \psi_d)$ is Haar-random on the unit sphere of $\mathbb{C}^d$.

A Haar-random unit vector can be generated by normalizing a complex Gaussian vector. Let

$$Z = (Z_1, \dots, Z_d), \quad (16)$$

where the Z_j are independent standard complex Gaussian variables, and set

$$|\psi\rangle = \frac{Z}{\|Z\|}. \quad (17)$$

This construction gives the Haar distribution because the joint density of Z is proportional to

$$\exp(-\|Z\|^2), \quad (18)$$

and therefore depends only on the norm $\|Z\|$. It is invariant under every unitary transformation $U \in U(d)$: the random vectors Z and UZ have the same distribution. Consequently $Z/\|Z\|$ and $UZ/\|Z\|$ have the same distribution on the unit sphere. By uniqueness of the unitarily invariant probability measure on that sphere, this distribution is precisely Haar measure. Then

$$|\psi_1|^2 = \frac{|Z_1|^2}{|Z_1|^2 + \dots + |Z_d|^2}. \quad (19)$$

By Eq. (6), each component can be written as $Z_j = (X_j + iY_j)/\sqrt{2}$, with X_j, Y_j independent $N(0, 1)$ variables. Therefore

$$|Z_j|^2 = \frac{X_j^2 + Y_j^2}{2}. \quad (20)$$

Since $X_j^2 + Y_j^2 \sim \chi_2^2$, and $\chi_2^2/2$ has the exponential distribution of mean one, $|Z_j|^2 \sim \text{Exp}(1)$, equivalently $|Z_j|^2 \sim \text{Gamma}(1, 1)$. The independence of the $|Z_j|^2$ follows from the independence of the pairs (X_j, Y_j) . To avoid overloading notation, write

$$E_j = |Z_j|^2, \quad (21)$$

with the E_j independent exponential random variables of mean one. Let

$$W = E_2 + \dots + E_d. \quad (22)$$

Then E_1 and W are independent, with $E_1 \sim \text{Gamma}(1, 1)$ and $W \sim \text{Gamma}(d-1, 1)$. Therefore

$$|\psi_1|^2 = \frac{E_1}{E_1 + W} \sim \text{Beta}(1, d-1). \quad (23)$$

This gives a useful symmetry with the real case below: the complex transition probability is beta distributed with parameters $(1, d-1)$, whereas the real case has parameters $(1/2, (d-1)/2)$.

For completeness, and also to make the tail explicit, we compute the same probability directly. For $0 \leq \varepsilon < 1$,

$$\begin{aligned} \mathbb{P}\{|\psi_1|^2 \geq \varepsilon\} &= \mathbb{P}\left\{\frac{E_1}{E_1 + W} \geq \varepsilon\right\} \\ &= \mathbb{P}\left\{E_1 \geq \frac{\varepsilon}{1-\varepsilon} W\right\} \\ &= \mathbb{E}\left[\exp\left(-\frac{\varepsilon}{1-\varepsilon} W\right)\right]. \end{aligned} \quad (24)$$

The last equality is just conditioning on W : since E_1 is independent of W and has exponential tail $\mathbb{P}\{E_1 \geq t\} = e^{-t}$, one has

$$\mathbb{P}\{E_1 \geq sW \mid W = w\} = e^{-sw}, \quad (25)$$

and averaging over W gives $\mathbb{E}(e^{-sW})$. Since W is the sum of $d-1$ independent exponential variables, $W \sim \text{Gamma}(d-1, 1)$, and its Laplace transform is

$$\mathbb{E}(e^{-sW}) = (1+s)^{-(d-1)}. \quad (26)$$

With $s = \varepsilon/(1-\varepsilon)$, this gives

$$\mathbb{P}\{|\psi_1|^2 \geq \varepsilon\} = (1-\varepsilon)^{d-1}. \quad (27)$$

At the endpoint $\varepsilon = 1$, the same formula gives zero; equivalently, a Haar-random vector is parallel to the fixed vector with probability zero for $d \geq 2$. The same formula also follows immediately from the cumulative distribution function of $\text{Beta}(1, d-1)$, since

$$\mathbb{P}\{T \geq \varepsilon\} = (1-\varepsilon)^{d-1}, \quad T \sim \text{Beta}(1, d-1). \quad (28)$$

Since $|\langle\phi|\psi\rangle|^2 = |\psi_1|^2$, Eq. (14) follows. Eq. (15) follows by putting $\varepsilon = \delta^2$, because δ is the amplitude-overlap threshold whereas ε is the corresponding Born-probability threshold, and then using $1-x \leq e^{-x}$. $\square$

Several consequences are immediate. The mean transition probability is

$$\mathbb{E}|\langle\phi|\psi\rangle|^2 = \frac{1}{d}, \quad (29)$$

which also follows from symmetry, since $\sum_{j=1}^d |\psi_j|^2 = 1$. Moreover, for every fixed threshold $\varepsilon > 0$,

$$\mathbb{P}\{|\langle\phi|\psi\rangle|^2 \geq \varepsilon\} \leq \exp[-(d-1)\varepsilon]. \quad (30)$$

Thus large transition probabilities between independently chosen random pure states are exponentially suppressed in the Hilbert-space dimension. In this precise sense, two random pure states are quasi-orthogonal with probability tending to one exponentially fast with d .

It is useful to separate two scales. The typical Born probability $|\langle\phi|\psi\rangle|^2$ is of order d^{-1} , while the typical amplitude $|\langle\phi|\psi\rangle|$ is of order $d^{-1/2}$. Thus random pure states are not exactly orthogonal, but their mutual transition probabilities are normally microscopic in large Hilbert spaces.

Exponential packing consequence

In a d -dimensional Hilbert space $\mathcal{H}_d$, the maximum number of exactly mutually orthogonal unit vectors is d . We relax exact orthogonality as follows. Two unit vectors $|\psi_i\rangle, |\psi_j\rangle \in \mathcal{H}_d$ are called ε -quasi-orthogonal if

$$|\langle\psi_i|\psi_j\rangle|^2 \leq \varepsilon. \quad (31)$$

A family $\{|\psi_i\rangle\}_{i=1}^M$ is ε -quasi-orthogonal if this inequality holds for every pair $i \neq j$.

Let $M_\varepsilon(d)$ denote the largest possible cardinality of such a family. The exact overlap tail above gives the following random-coding lower bound:

$$M_\varepsilon(d) \geq \left\lfloor \exp\left[\frac{d-1}{2}(-\log(1-\varepsilon))\right] \right\rfloor. \quad (32)$$

Indeed, sample M unit vectors $|\psi_1\rangle, \dots, |\psi_M\rangle \in \mathcal{H}_d$ independently from Haar measure. For any pair $i \neq j$, Eq. (14) gives

$$\mathbb{P}(|\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) = (1-\varepsilon)^{d-1}. \quad (33)$$

By a union bound over the $\binom{M}{2} < M^2/2$ pairs,

$$\mathbb{P}(\exists i \neq j : |\langle\psi_i|\psi_j\rangle|^2 \geq \varepsilon) < \frac{1}{2}M^2(1-\varepsilon)^{d-1}. \quad (34)$$

Choosing

$$M = \left\lfloor \exp\left[\frac{d-1}{2}(-\log(1-\varepsilon))\right] \right\rfloor \quad (35)$$

gives

$$\frac{1}{2}M^2(1-\varepsilon)^{d-1} \leq \frac{1}{2} < 1, \quad (36)$$

so the bad event cannot have probability one. Hence there exists at least one ε -quasi-orthogonal family of cardinality M , proving Eq. (32).

For small fixed ε , since $-\log(1-\varepsilon) = \varepsilon + O(\varepsilon^2)$, this implies

$$M_\varepsilon(d) \geq \exp(c\varepsilon d) \quad (37)$$

for some constant $c > 0$, fixed $\varepsilon > 0$, and sufficiently large d . The statement should be understood in the fixed- ε , large- d regime; no bound of the form $M_\varepsilon(d) \leq \exp(C\varepsilon d)$ can hold uniformly down to $\varepsilon = O(1/d)$, because an orthonormal basis already gives $M_\varepsilon(d) \geq d$ for every $\varepsilon \geq 0$. Standard spherical-code bounds also give exponential-in- d upper bounds for fixed $\varepsilon > 0$ [13, 14], but the precise optimal exponent is not needed here. We therefore use only the random-coding lower bound.

For a system of n qubits, $d = 2^n$. Hence, for fixed $\varepsilon > 0$,

$$M_\varepsilon(2^n) \geq \exp(c\varepsilon 2^n), \quad (38)$$

which is doubly exponential in the number n of qubits. Thus the Hilbert space of a macroscopic environment has vastly more than enough kinematic room to accommodate an enormous number of mutually quasi-orthogonal environmental records.

It is important not to overinterpret this packing statement. The bound concerns pairwise transition probabilities between pure states. It does not imply that a d -dimensional environment can store exponentially many reliably distinguishable classical records. If $M \gg d$, the packed states are necessarily linearly dependent, and pairwise small overlaps alone do not guarantee a measurement that recovers the corresponding labels with high average success probability. The relevance for decoherence is more modest and more appropriate: for a fixed, moderate, or dynamically selected set of macroscopic alternatives, the suppression of pairwise overlaps is the quantity that controls the off-diagonal terms of the reduced state. In that setting, high-dimensional Hilbert space provides ample kinematic room for environmental branch states to become effectively orthogonal.

This statement is closely related in spirit to the concentration phenomenon underlying the Johnson–Lindenstrauss lemma, but the emphasis is different. Johnson–Lindenstrauss is a finite-set embedding result: N vectors can be randomly projected to $O(\varepsilon^{-2} \log N)$ dimensions while preserving pairwise distances up to relative error ε . The present argument uses the complementary viewpoint. In a fixed high-dimensional Hilbert space, independently chosen pure states already have Born transition probabilities of order $1/d$, and large overlaps are exponentially suppressed. Thus Johnson–Lindenstrauss concerns the preservation of geometry under random dimensional reduction, whereas the present use concerns the generic near-orthogonality already available in high dimension.

Real comparison

The complex case is the natural one for quantum mechanics. For comparison, the real case has the same concentration behaviour, but a slightly less elementary closed form. Let $u, v \in \mathbb{R}^d$ be independent Haar-random unit vectors on the real sphere S^{d-1} . By orthogonal invariance, fix $v = e_1$. A Haar-random real unit vector is generated analogously by normalizing a standard real Gaussian vector: if $G = (G_1, \dots, G_d)$, where the G_j are independent $N(0, 1)$ variables, then $G/\|G\|$ is invariant under all orthogonal transformations and hence is uniform on S^{d-1} . Thus, for $u = G/\|G\|$,

$$|\langle v, u \rangle|^2 = \frac{G_1^2}{G_1^2 + \dots + G_d^2}. \quad (39)$$

Here $G_1^2 \sim \text{Gamma}(1/2, 2)$, and $G_2^2 + \dots + G_d^2 \sim \text{Gamma}((d-1)/2, 2)$, in the same shape-scale convention. The scale is now 2 rather than 1, but it is common

to numerator and denominator and therefore cancels in the ratio. The beta-ratio rule gives

$$|\langle v, u \rangle|^2 \sim \text{Beta} \left(\frac{1}{2}, \frac{d-1}{2} \right). \quad (40)$$

Equivalently, its density on $(0, 1)$ is

$$f_d(t) = \frac{t^{-1/2}(1-t)^{(d-3)/2}}{B(\frac{1}{2}, \frac{d-1}{2})}. \quad (41)$$

Thus

$$\mathbb{P}\{|\langle v, u \rangle|^2 \geq \varepsilon\} = \int_{\varepsilon}^1 \frac{t^{-1/2}(1-t)^{(d-3)/2}}{B(\frac{1}{2}, \frac{d-1}{2})} dt. \quad (42)$$

For fixed $\varepsilon > 0$, this again decays exponentially in d , because the factor $(1-t)^{(d-3)/2}$ is exponentially small throughout the integration range $t \geq \varepsilon$. More explicitly,

$$\begin{aligned} \int_{\varepsilon}^1 t^{-1/2}(1-t)^{(d-3)/2} dt &\leq \varepsilon^{-1/2} \int_{\varepsilon}^1 (1-t)^{(d-3)/2} dt \\ &= \frac{2\varepsilon^{-1/2}}{d-1} (1-\varepsilon)^{(d-1)/2}. \end{aligned} \quad (43)$$

Furthermore, $B(1/2, (d-1)/2)^{-1} = O(\sqrt{d})$ by Stirling's formula. Hence, for each fixed $\varepsilon \in (0, 1)$,

$$\begin{aligned} \mathbb{P}\{|\langle v, u \rangle|^2 \geq \varepsilon\} &\leq C_{\varepsilon} d^{-1/2} (1-\varepsilon)^{(d-1)/2} \\ &\leq C_{\varepsilon} d^{-1/2} \exp \left[-\frac{(d-1)\varepsilon}{2} \right] \end{aligned} \quad (44)$$

for a constant C_{ε} independent of d . The exponent is roughly half the complex exponent, reflecting the fact that $\mathbb{C}^d$ has twice as many real degrees of freedom as $\mathbb{R}^d$. The qualitative high-dimensional conclusion is nevertheless the same: a random real unit vector is almost orthogonal to an independently chosen one with probability tending to one exponentially fast in the dimension.

Relation to Lévy's lemma

The preceding argument is self-contained and uses no spherical isoperimetric black box. It exploits the special observable relevant to pure-state quantum transition probabilities, namely

$$|\psi\rangle \mapsto |\langle \phi | \psi \rangle|^2. \quad (45)$$

Lévy's lemma is the corresponding general concentration theorem: it replaces this particular transition probability by an arbitrary Lipschitz observable on the high-dimensional unit sphere. In that broader setting one obtains bounds of the schematic form

$$\mathbb{P}\{|f - \text{typical value}| \geq \eta\} \lesssim \exp \left[-c \frac{d\eta^2}{L^2} \right], \quad (46)$$

where L is the Lipschitz constant of f . Proving this general statement requires the usual concentration-of-measure machinery, for example spherical isoperimetry.

The transition-probability observable itself is Lipschitz. If $P_{\phi} = |\phi\rangle\langle\phi|$ and $f(\psi) = \langle\psi|P_{\phi}|\psi\rangle = |\langle\phi|\psi\rangle|^2$, then for two unit vectors ψ, ψ' ,

$$|f(\psi) - f(\psi')| \leq 2 \|\psi - \psi'\|. \quad (47)$$

Thus f has Lipschitz constant at most 2 with respect to Euclidean chordal distance, and hence also with respect to geodesic distance up to the usual constant convention. Applying Lévy's lemma to this particular observable would therefore prove concentration, but with a generic Gaussian tail in the deviation parameter. The exact beta-distribution calculation above is sharper for transition probabilities: for fixed ε , it gives an exponent linear in ε , namely $\exp[-(d-1)\varepsilon]$, rather than the weaker generic Lipschitz-type behaviour quadratic in the deviation. For the quantum quasi-orthogonality claim above, no black-box input is needed: the exact distribution of the Born transition probability already gives the desired exponential suppression.

III. APPLICATION TO THE MEASUREMENT PROCESS

Consider a system $\mathcal{S}$ coupled to an apparatus-environment $\mathcal{E}$ initialised in $|E_0\rangle$:

$$|\Psi(0)\rangle = \left(\sum_i c_i |s_i\rangle \right) \otimes |E_0\rangle. \quad (48)$$

A measurement-like unitary correlates pointer outcomes with environmental records:

$$|\Psi(t)\rangle = \sum_i c_i |s_i\rangle \otimes |E_i(t)\rangle. \quad (49)$$

The reduced density matrix on $\mathcal{S}$ is

$$\rho_{\mathcal{S}}(t) = \sum_{i,j} c_i c_j^* \langle E_j(t) | E_i(t) \rangle |s_i\rangle \langle s_j|. \quad (50)$$

Off-diagonal coherences are weighted by the environmental overlaps $\langle E_j(t) | E_i(t) \rangle$, and are suppressed in modulus by at most $\sqrt{\varepsilon}$ whenever the branch states are ε -quasi-orthogonal.

The kinematic claim.

Throughout this section we adopt natural units $k_B = \hbar = 1$, so that entropies are dimensionless and inverse temperatures have units of time. If the branch states $\{|E_i(t)\rangle\}_{i \neq j}$ are well-modeled as a typical pair on the relevant high-dimensional subspace of $\mathcal{H}_{\mathcal{E}}$, then

$$\mathbb{E} [|\langle E_j(t) | E_i(t) \rangle|^2] = \frac{1}{d_{\text{eff}}}, \quad (51)$$

with d_{eff} the dimension of the subspace explored by the dynamics. For a generic chaotic many-body environment the accessible subspace is the microcanonical shell, so

$$d_{\text{eff}} \sim e^S, \quad S \sim n, \quad (52)$$

with S the (dimensionless) microcanonical entropy of the relevant environmental sector and n the number of constituent degrees of freedom; in conventional units the right-hand sides of (52) read e^{S/k_B} and nk_B , respectively. Equation (52) ties the geometric parameter d_{eff} directly to standard thermodynamics: the mean-square overlap is exponential in the entropy of the environment, so the typical overlap amplitude is of order $e^{-S/2}$.

The relevant dimension is not necessarily the full Hilbert-space dimension of $\mathcal{H}_{\mathcal{E}}$, but the dimension of the subspace $\mathcal{H}_{\text{acc}}$ accessible to the dynamics from the initial state on the timescale considered. Energy conservation, conserved charges, and finite-time evolution can all restrict the accessible subspace. For a many-body environment with Hamiltonian $H_{\mathcal{E}}$ and an initial state localized within an energy window $[E, E + \Delta E]$, the accessible subspace is often well approximated by the microcanonical shell, with

$$d_{\text{eff}} = \dim \mathcal{H}_{\text{acc}} \sim e^{S(E)/k_B}. \quad (53)$$

In the natural units used here, this is simply $d_{\text{eff}} \sim e^{S(E)}$. For chaotic many-body dynamics with bounded local interactions, $S(E)$ is extensive in the number of constituents, so d_{eff} remains exponential in system size. For integrable or many-body-localized environments, additional conserved quantities can drastically reduce d_{eff} ; the geometric reservoir is then correspondingly smaller.

This is vastly stronger than needed to render coherences operationally invisible on any timescale shorter than the Poincaré recurrence time, which for a generic chaotic many-body system is astronomically large—heuristic arguments suggest a scaling at least as $\exp(d_{\text{eff}}) \sim \exp(e^S)$, but the precise value is model-dependent and irrelevant to our argument [15, 16].

Minimal dynamical input.

The argument is kinematic rather than dynamical. It assumes that the interaction Hamiltonian has already selected the relevant pointer alternatives and that the associated environmental branch states explore a sufficiently large accessible subspace in a typical manner. It does not by itself derive the pointer basis, nor does it prove thermalization or equilibration. Rather, it explains why, once distinct branches are driven into a large environmental Hilbert space, their mutual overlaps are generically expected to be extremely small.

In a measurement-like interaction, after a pointer basis has been selected, one may write schematically

$$|E_i(t)\rangle = U_i(t)|E_0\rangle, \quad (54)$$

where the conditional unitary $U_i(t)$ depends on the pointer value i . The geometric argument does not prove that the vectors $U_i(t)|E_0\rangle$ are Haar-random. The required dynamical assumption is only the weaker pair-typicality condition that, for $i \neq j$, their overlap statistics are close to those of two typical vectors in the accessible subspace $\mathcal{H}_{\text{acc}}$:

$$|\langle E_j(t)|E_i(t)\rangle|^2 \sim d_{\text{eff}}^{-1}, \quad d_{\text{eff}} = \dim \mathcal{H}_{\text{acc}}. \quad (55)$$

Chaotic, nonintegrable environments provide the usual physical reason to expect such typicality after a suitable relaxation or decoherence time. Integrable, many-body-localized, weakly coupled, or strongly constrained environments may fail to satisfy it. Thus the geometry supplies the available reservoir of nearly orthogonal records, while the Hamiltonian must still populate that reservoir. This assumption is in the same broad family as the typicality assumptions used in canonical typicality and quantum thermalization [17–21].

Limitations.

Two limitations should be kept explicit. First, quasi-orthogonality does not select the pointer basis. The preferred basis is determined by the structure of the system–environment interaction, not by high-dimensional geometry alone.

Second, small off-diagonal terms do not by themselves turn an improper mixture into a proper one. The geometric estimates therefore strengthen the FAPP suppression of interference, but they do not resolve the interpretive core of the measurement problem.

Third, pairwise suppression should not be conflated with arbitrary classical readout capacity. If the number of alternatives is so large that the full Gram matrix is densely populated by small but numerous off-diagonal terms, pairwise bounds alone need not control all global decoding or distinguishability questions. The decoherence application therefore concerns fixed, moderate, or dynamically selected families of macroscopic alternatives, not an exponentially large alphabet of perfectly readable records.

A strict, rather than merely FAPP, breakdown of unitary equivalence between macroscopic outcomes can be obtained in the infinite tensor product limit through sectorization and factorization [22–25], of which the present finite-dimensional, geometric estimates may be viewed as a quantitative companion: exponentially small (rather than exactly vanishing) overlaps between macroscopically distinct environmental records.

Schrödinger’s “jellification” worry.

Schrödinger’s worry was that, if unitary quantum mechanics applies universally, macroscopically distinct alternatives should blur into a single “featureless jelly.” In

the present language, such blurring would require appreciable overlaps between the environmental records correlated with different macroscopic alternatives. Under the pair-typicality assumption above, however,

$$|\langle E_j(t)|E_i(t)\rangle|^2 \sim d_{\text{eff}}^{-1} \sim e^{-S},$$

so the typical overlap amplitude is of order $e^{-S/2}$. For a macroscopic environment this is FAPP indistinguishable from zero. The branches do not blur because the accessible Hilbert space contains an enormous supply of nearly orthogonal directions. This answers the jellification worry operationally, while leaving the deeper interpretive questions untouched.

IV. DISCUSSION

The geometric perspective separates two ingredients in decoherence. The first is dynamical: the interaction Hamiltonian selects a pointer basis and correlates distinct pointer values with distinct environmental records. The second is kinematic: once those records behave like typical vectors in a high-dimensional accessible subspace, their mutual overlaps are automatically tiny.

For an accessible subspace of dimension $d_{\text{eff}} \sim e^S$, the typical squared overlap is $d_{\text{eff}}^{-1} \sim e^{-S}$, and the typical overlap amplitude is $e^{-S/2}$. This suppression is not a new dynamical mechanism; it is the high-dimensional geometry underlying the effectiveness of environmental decoherence. The geometry explains why there is ample Hilbert-space room for many macroscopic records to coexist without appreciable interference, but it does not explain why a particular Hamiltonian produces those records, nor does it select a preferred basis.

This distinction also clarifies where the framework can fail. If the environment is integrable, many-body localized, weakly coupled, or restricted by additional conserved quantities, the branch states may not be typical in the relevant subspace. In that case the geometric reservoir exists but is not dynamically populated.

The argument is close in spirit to canonical typicality, where concentration of measure is used to show that most pure states of a large composite system have nearly thermal reduced states on small subsystems [17–19]. Here

the object of interest is different: the pairwise overlap between distinct environmental records. Infinite-tensor-product approaches to measurement obtain exact orthogonality between macroscopic sectors in the limit $n = \infty$; the present analysis can be read as the finite- n version, where exact sectorization is replaced by exponentially small overlap.

V. CONCLUSION

We have isolated a kinematic contribution to the suppression of macroscopic interference. In a d -dimensional Hilbert space, random pure states are almost orthogonal: for fixed $|\phi\rangle$ and Haar-random $|\psi\rangle$,

$$\mathbb{E}|\langle\phi|\psi\rangle|^2 = \frac{1}{d}, \quad \mathbb{P}(|\langle\phi|\psi\rangle|^2 \geq \varepsilon) = (1 - \varepsilon)^{d-1}.$$

Moreover, for fixed $\varepsilon > 0$, random coding produces ε -quasi-orthogonal families of size at least $\exp(c\varepsilon d)$. For an environmental subspace with $d_{\text{eff}} \sim e^S$, this means that typical squared branch overlaps are of order e^{-S} .

These facts do not replace decoherence dynamics. The Hamiltonian must still select the pointer basis and drive environmental records into typical regions of the accessible subspace. Nor do the estimates convert an improper mixture into a proper one. What the geometry does provide is a quantitative explanation of why, once environmental records have formed, macroscopic alternatives can coexist without producing Schrödinger’s “featureless jelly”: high-dimensional Hilbert space supplies an enormous reservoir of nearly non-interfering directions.

ACKNOWLEDGMENTS

This text was partially created and revised with assistance from large language models. This research was funded in whole or in part by the Austrian Science Fund (FWF) Grant DOI: 10.55776/PIN5424624. The author acknowledges TU Wien Bibliothek for financial support through its Open Access Funding Programme.

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